

Lab One: Introduction

Through the completion of this lab, you should be able to:

- understand and practice the use of Pseudocode for algorithmic design;
- set up your system or workspace for using Python 3 for the semester;
- practice the creation of a very simple Python 3 program.

The above statements are the *learning outcomes* of this laboratory and will be achieved in concert with the other learning activities that you undertake for this unit.

Task One: Algorithmic Design

1. Firstly, your tutor will briefly introduce themselves, spend some time explaining how the seminars (labs) will work this semester and go around the room to get you to introduce yourselves.
2. Utilising a piece of paper and a pen or if you prefer, digitally using your own device, write separate algorithms as Pseudocode to achieve the below (i.e. solve the following problems):
 - a. Ask the user to input two numbers, add them together and output (display) the result;
 - b. Ask the user to input two numbers, multiply them together and output (display) the result;
 - c. Ask the user to input a number, multiply it by itself and output (display) the result;
 - d. Ask the user to input a number, subtract 51 from it and output (display) the result.

Task Two: Workspace Setup

3. Decide which of the two (really, three) ways you wish to work with the lab content this semester:
 - a. the UniApps streaming application system on your own laptop computer (this method is generally recommended for most people);
 - b. the UniApps streaming application system on the lab machines (the least preferred method of the three) or;
 - c. installing the required programs for this unit on your own computer system (if they are already installed for another reason such as for another unit, it is recommended to go down this path instead to avoid any conflicts).
4. Undertake the steps below depending on the decision that you've made above:
 - a. *if UniApps on your own laptop computer:* visit the UniApps website in your browser at <https://uniapps.uwa.edu.au/> and then login with your UWA credentials. Follow the prompts to download and install the "UniApps" software for your computer (either Mac or Windows). Once installed, navigate to the "UniApps List" from the LMS site for this unit and click the "Launch" button under "IDLE". You may have to do this clicking a couple of times to get it working properly on your machine – that is to be expected;
 - b. *if UniApps on the lab machines:* navigate to the "UniApps List" from the LMS site for this unit and click the "Launch" button under "IDLE". You may have to do this clicking a couple of times to get it working properly on your machine – that is to be expected;
 - c. *if 'Your Own' computer system:* you will need to install the latest version of Python 3 from the Python website, using your web browser, at <https://python.org/> and following the prompts to download the installer for your system and then running the installer. Ensure that 'IDLE' is installed as part of this process. Then, open up IDLE using the usual

process on your system (i.e. Start Menu / Search on a modern Windows machine or the Applications folder / Launchpad on a Mac) once it has been installed.

Task Three: Python Program Creation

5. Utilising the tools of your choosing, write the Pseudocode for a program that simply and only outputs the phrase "Hello, World!" (without the quotes).
6. Either on your own computer or the lab machines, depending on what you chose above, open up the "IDLE" program (it is probably still open from Task Two) utilising the method discussed in Task Two. This is how you will access IDLE for any future tasks, so remember how you did it.
7. Utilise IDLE to create a new program named **HelloWorld.py** and within this program, implement the program as described above (i.e. write a program to output the phrase).
8. Run the program and ensure that the program runs as described. Correct any errors in your code, if identified and hence required, once you have run and hence tested the program.

End of Lab One.